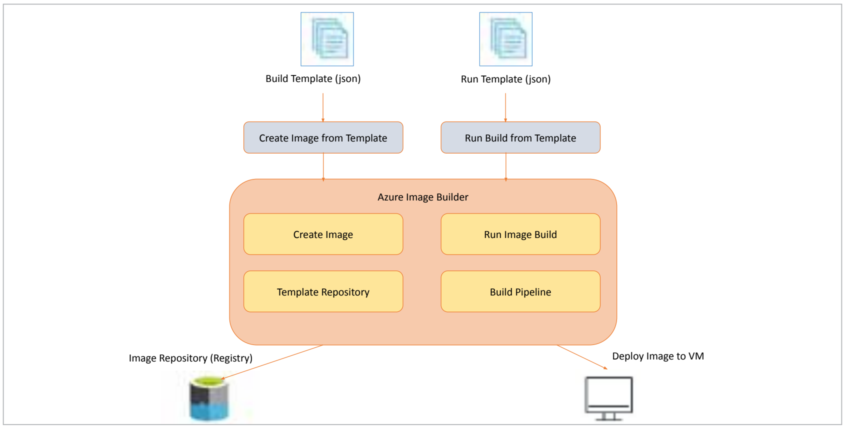

Figure 3: Azure Image Builder

template. This can be a JSON (to export/reuse from native scripts) or YAML script based simplified solution that can be integrated with any cloud service provider's native deployment pipeline or third-party pipeline like Ansible to enable infrastructure provisioning activities on-the-go with flexibility.

It is easy to do large scale infrastructure creation, spin-up development, test instances during the implementation cycle or create pre-production or production instances during horizontal scaling, with Terraform scripts. These can be run with resource schedulers, and can create network configuration or secured environment setups for software defined networks (SDN) using Terraform templates.

Terraform scripts can be used to spin-up instances or to destroy environments for disposable temporary test environments in a cloud platform. Terraform templates can be validated before actual provisioning. Terraform workflow has three steps — write (create Terraform templates for IaC), plan (preview the changes before applying them in the cloud environment to validate) and apply (to provision infrastructure quickly).

For a cloud-agnostic solution like multi-cloud architecture, it is easy to adapt Terraform templates for IaC to quickly create a uniform setup and configuration for multiple environments, reduce complexity in creating multiple parallel scripts, and reuse these scripts as much as possible for each cloud service provider in the multi-cloud setup.

Some claims on the Internet, though, state that Terraform is not cloud-agnostic but a wrapper running on top of the native cloud services of any cloud service provider.

Cloud-agnostic provisioning using Terraform

Infrastructure as Code (IaC) helps to look at infrastructure as a software module and programmatically handle it through scripts. This can be quite useful for cloud platforms to quickly provision virtual infrastructure components like compute, storage, network, security and databases.

Infrastructure as Code (IaC) templates can be classified as cloud native templates (like AWS CloudFormation, Azure Resource Manager and Google Deployment Manager templates), and cloud-agnostic templates (like Terraform templates).

Cloud native templates are flexible and customisable to the native platform, and can be integrated with any native service easily. Cloud-agnostic templates are suitable for multi-cloud application services in order to have uniform infrastructure scripts across cloud service providers to reduce rework in IaC script activities. For example, if you want to deploy a Web application and API services with disaster recovery services (active-passive deployment) in the AWS platform, it would be better to choose cloud-agnostic templates.

Google Cloud Platform (GCP) has a native template service called Cloud Foundation Toolkit (CFT), which contains a set of reference templates that can be used as is or customised to your IaC needs. The reference templates available in CFT are deployment manager templates (GCP native) or Terraform templates (cloud-agnostic). They are simple Python scripts that can be customised easily.

Other popular cloud native IaC facilities

Infrastructure as Code as well as cloud automation for provisioning and virtualisation requirements are very helpful to accelerate build, deployment, templatizing and cloud management activities. They are even more helpful when it comes to multi-cloud management, as preparing a provisioning script for each cloud is inconsistent and inefficient. Infrastructure as Code helps to automate this across different cloud environments with ease.

Though cloud native facilities like CloudFormation in AWS, ARM templates in Azure and CDM templates in GCP help to achieve this Infrastructure as Code, they are mostly native to the cloud platform and don't have the agility and flexibility for multi-cloud management. On the other hand, Google Anthos, HashiCorp Terraform, and Red Hat Ansible templates can help in multi-cloud management, enabling the development of zero-touch deployment scripts and provisioning templates for cloud tower setup.

There are various VM image building tools like Packer and Terraform, but Azure Image Builder